

Note: Auto-demoted to one-compartment ($\Delta AICc < 2$ vs 1c and/or poor rate separation or tiny B, or $n < 6$).

Subject S1

Study Metadata

Field	Value
study_id	UserStudy1
species	Mus musculus
route	IV bolus
dose	100
model	two
time_units	h
conc_units	a.u.
allow_demote	True
k10	0.35
k12	0.6
k21	0.3
V1	0.04
V2	0.06
subject	S1

Data Summary

n	42
Time span (min → max)	0.00 → 24.00 h
C_max (observed)	4.39 a.u.
Global $t_{1/2}$ (log-linear)	1.40 h
$ m_{\text{early}} / m_{\text{late}} $ (log-slopes)	1.04

Fit Results (with 95% CI)

Parameter	Estimate	95% CI
Vd	22.5	[22.2, 22.8]
kel	0.488	[0.474, 0.501]
CI	11	[10.7, 11.2]

t_half	1.42	[1.38, 1.46]
C0	4.45	[4.39, 4.5]
AUC	9.12	[8.91, 9.33]
MRT	2.05	[1.99, 2.11]

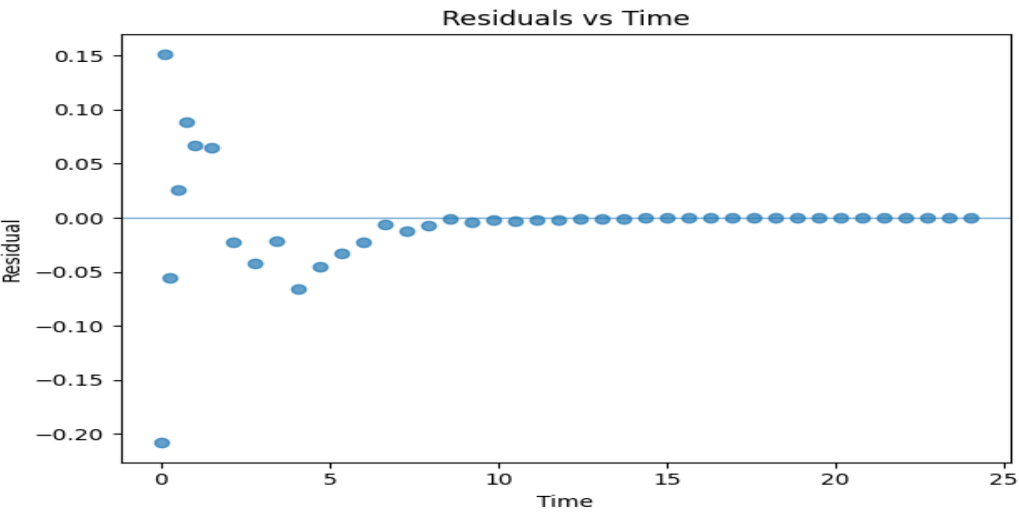
Goodness-of-Fit

R²	0.999
AIC	-250.8
AICc	-250.4

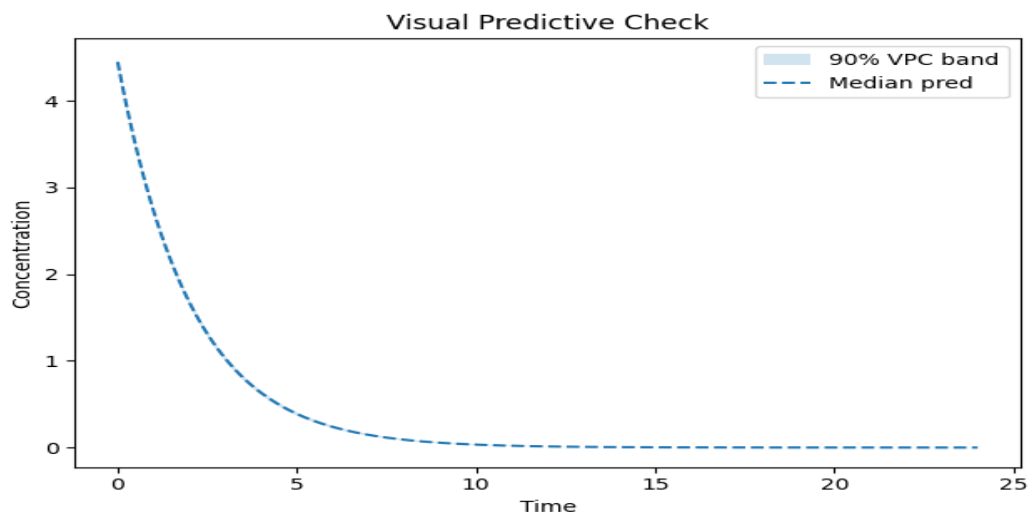
Model Selection Diagnostics

AICc (1c)	-250.45
AICc (2c)	-245.67
ΔAICc (2c – 1c)	4.77
sep_rel (2c)	0.000
B_rel (2c)	0.255
Demoted	Yes

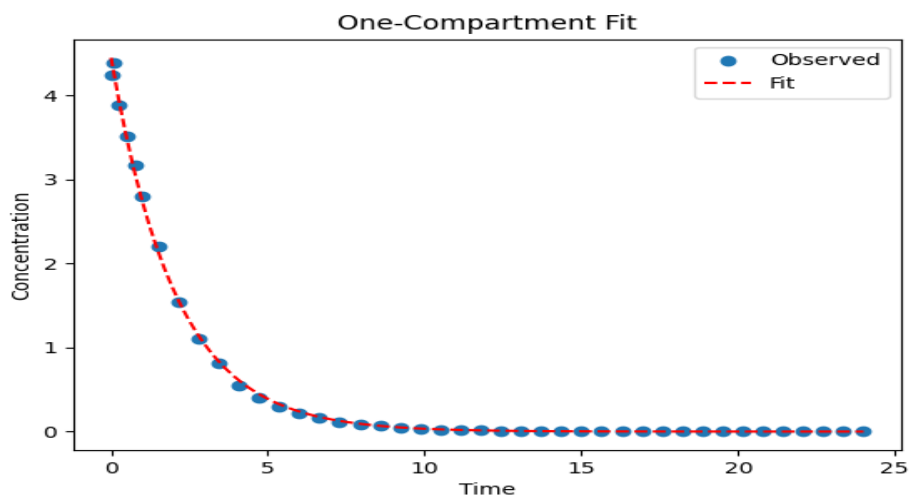
Residual Diagnostics



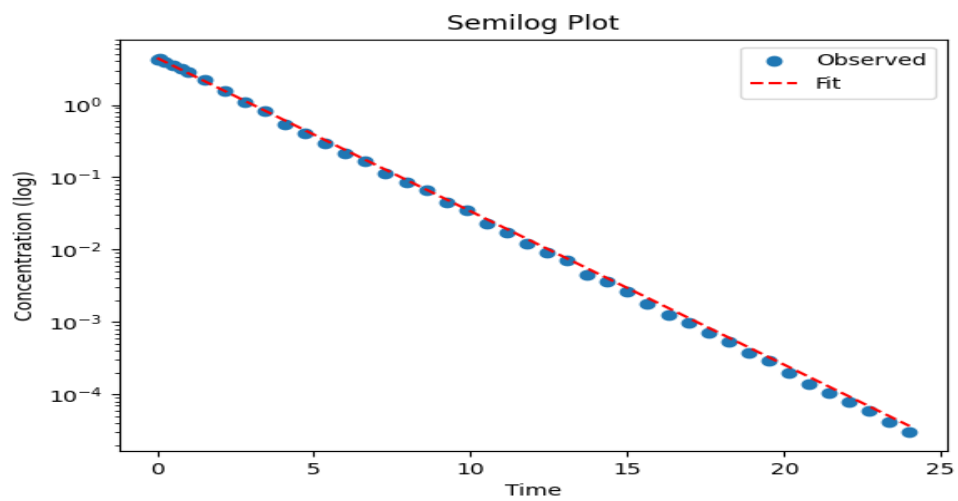
Visual Predictive Check (5–95%)



Linear Fit



Semilog Fit



Dosing Timeline

